# Web Based ML Model Evaluator

**Team: 13**

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**Functional Requirements**

Data Ingestion and Preprocessing:

- **FR-1 Dataset Upload:** User should be able to upload their dataset file.
- **FR-2 Dataset Preview:** System should be able to analyze the uploaded CSV and present a dataset summary showing total rows, column names, data types, missing value counts, and a 5 row data preview.
- **FR-3 Preprocessing:** System should be able to preprocess and clean the dataset such as handling missing values.

ML Task Routing & Configuration Module:

- **FR-4 Task Selection:** System should be able to allow the user to select an ML problem type (Classification, Regression, etc) before configuring models.
- **FR-5 Feature & Target Selection (Supervised):** For Classification and Regression tasks, the system should allow the user to select one column as the target variable and one or more columns as features.
- **FR-6 Feature Selection & Hyperparameter Input (Clustering):** For clustering, the system should allow the user to input the number of clusters.
- **FR-7 Train/Test Partitioning:** For Supervised tasks, the system shall provide an option to allow users to adjust the train/test split ratio.

Execution & Evaluation Module

- **FR-8 Task-Specific Algorithm Execution:** System shall execute model fitting based on user selection:
    - Classification: Logistic Regression, Decision Tree Classifier, Random Forest Classifier, K-Nearest Neighbors (KNN).

    - Regression: Linear Regression, Decision Tree Regressor, Random Forest Regressor.
    - Clustering: K-Means Clustering.
- **FR-9 Classification Performance Metrics:** System should compute and display Accuracy, Precision, Recall, F1-Score, and a confusion matrix if needed.
- **FR-10 Regression Performance Metrics:** System should compute and display mean absolute error, Mean Squared Error (MSE), Root Mean Squared Error and R2 score.
- **FR-11 Clustering Performance Metrics & Visuals:** System should compute Silhouette Score and Davies-Bouldin Index, and display a 2D Scatter Plot generated via Principal Component Analysis (PCA) showing cluster assignments.
- **FR-12 Model Comparison Dashboard:** System should allow users to compare evaluation metrics of two or more trained models side by side.
- **FR-2 Robust Error Handling & Data Validation:** The backend should intercept invalid execution states and return structured, human-readable error messages to the UI.

**Non-Functional Requirements**

- **NFR-1 Pipeline Response Latency:** Ingestion, feature scaling, model fitting, and metric generation for datasets under 10 MB shall execute within 30 seconds per run.
- **NFR-2 Ease of Use:** The web interface should guide the user through a clear process [upload -> select columns -> train model -> display results]
- **NFR-3 Browser Compatibility:** The web application should run smoothly on modern web browsers (Google Chrome, Firefox, Microsoft Edge) without requiring any extra browser plugins.
- **NFR-4 File Limit:** The system should restrict file uploads to a maximum size to prevent server slowdowns.